ECO102:Introduction to Macroeconomics

Lecture 3: Monitoring Jobs and Inflation

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Outline

- Employment and Unemployment
- Key Definitions
- Labour Market Indicators
- Types of Unemployment
- “Natural” Unemployment
- Full Employment
- Full Employment and Potential GDP
- Price Level, Inflation and Deflation

Employment and Unemployment

- *Arguably the most important market within the economy: The labour Market*
- *Source of the largest component of GDP*
- *A key step towards poverty reduction: job creation*
- *Two key components: i) Employment and ii) Unemployment*

- *Why is Unemployment a Problem?*
  - *Lost incomes and production*

    Lost income → Lower capacity for consumption

    Lost Production → Less Product available for consumption → Lower investment in capital → Lower standard of Living (Both present and future)
  - *Lost human capital*

Key Definitions

- The U.S. Census Bureau divides the population into a few categories that allows them to describe the anatomy of the labour market more comprehensively

- ***Working-age Population:*** The total number of people aged 16 years or older and who are not in jail, hospital, or some other form of institutional care;

  Divided into two categories: i) Those in the labour force, and ii) those not in the labour force

- ***Labour Force:*** Sum of the employed and unemployed people

$$Labour\ Force = Number\ of\ People\ Employed + Number\ of\ People\ Unemployed$$

Key Definitions

- ***Employed:*** Any person who holds a part-time or a full-time job is considered to employed

- ***Unemployed:*** Any person who is available for work and falls under one of these three categories:
  - Without work but has made specific efforts to find a job within the previous four weeks
  - Waiting to be called back to a job from which he or she has been laid off
  - Waiting to start a new job within 30 days

Key Definitions

POPULATION LABOUR FORCE CATEGORIES

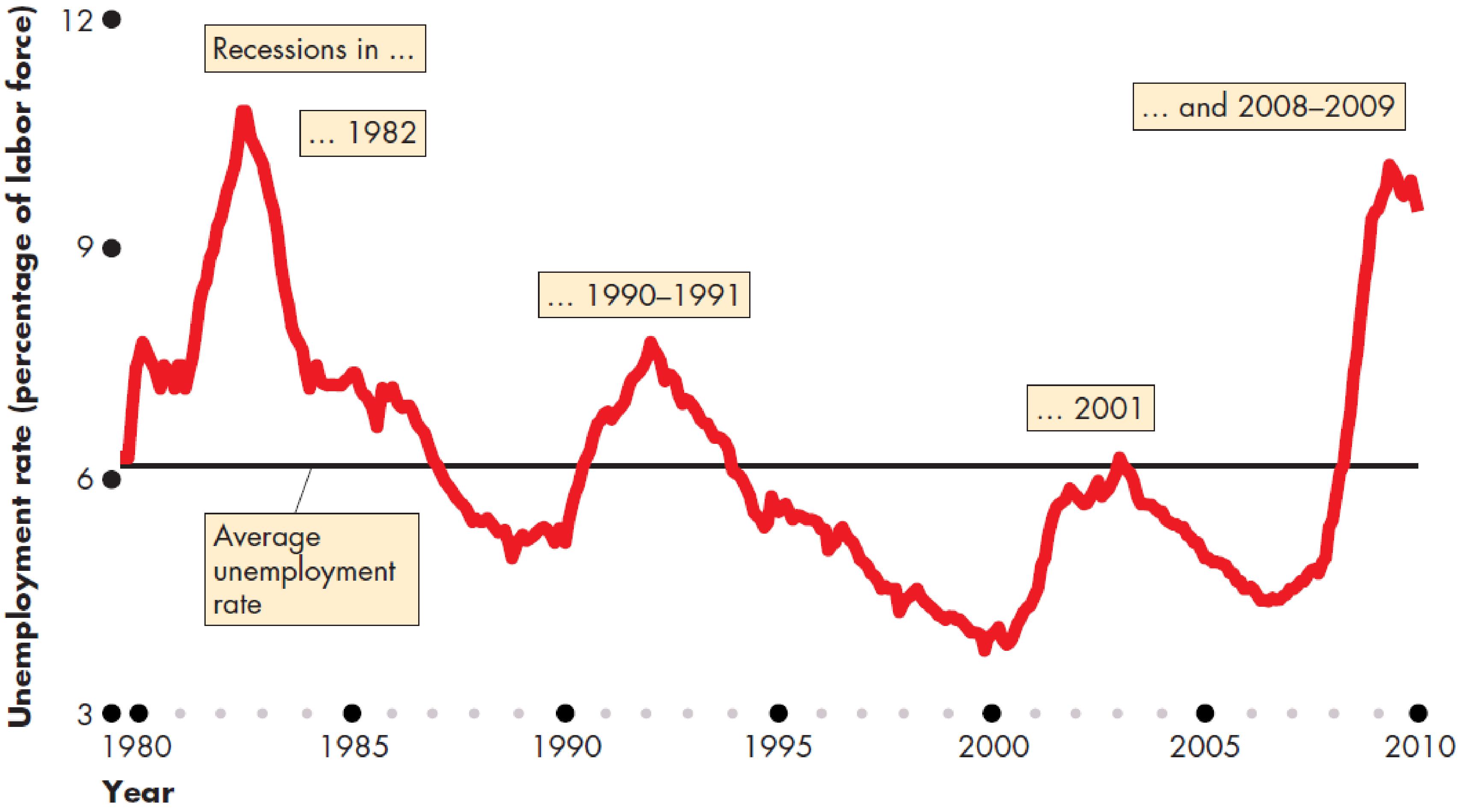
Labour Market Indicators

- ***The Unemployment Rate:*** The unemployment rate is the percentage of the people in the labour force who are unemployed

$$Unemployment\ Rate = \frac{Number\ of\ People\ Unemployed}{Labour\ Force} \times 100$$

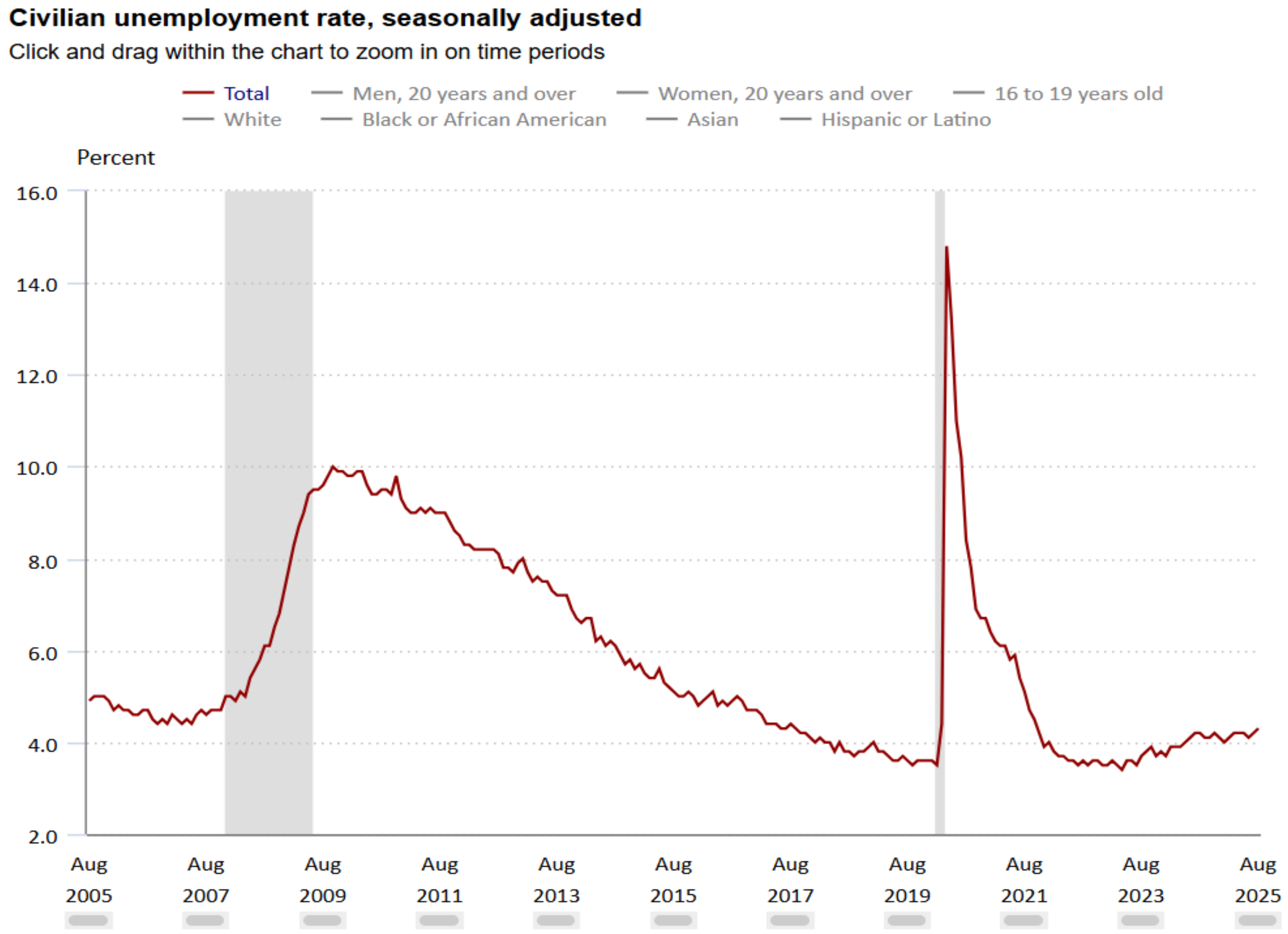
Question: Assume for a given country and on a given year, the number of people employed is 140 million and the number of people unemployed is 15 million. What is the unemployment rate for that country during that year?

Labour Market Indicators



Source of data: Bureau of Labor Statistics.

Labour Market Indicators



Labour Market Indicators

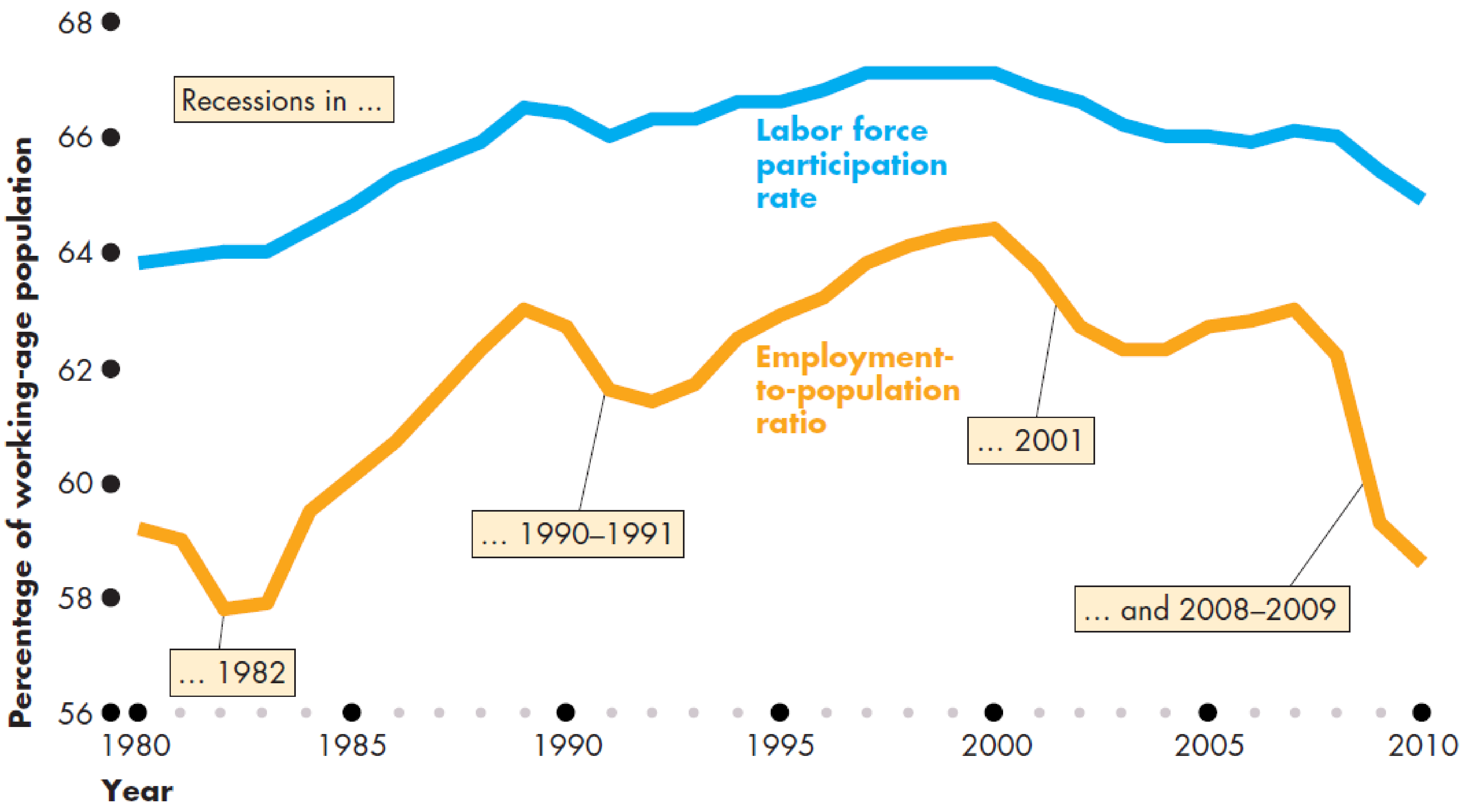
- ***Employment-to-population Ratio:*** The percentage of working age people who have jobs.

$$Employment - to - population\ Ratio = \frac{Number\ of\ People\ Employed}{Working - age - population} \times 100$$

- ***Labour Force Participation Rate:*** The percentage of the working-age population who are members of the labour force

$$Labour\ Force\ Participation\ Rate = \frac{Labour\ Force}{Working - age - population} \times 100$$

Labour Market Indicators



Source of data: Bureau of Labor Statistics.

Types of Unemployment

- ***Frictional Unemployment:*** The unemployment that arises from the normal **labour turnover**, from people entering and leaving the labour force and from the ongoing creation and destruction of jobs, is called frictional unemployment

It is a permanent and healthy phenomenon in a dynamic and growing economy

Types of Unemployment

- ***Structural Unemployment:*** The unemployment that arises when changes in technology or international competition change the skills needed to perform jobs or change the locations of jobs is called structural unemployment

Structural unemployment usually lasts longer than frictional unemployment because workers must retrain and possibly relocate to find a job

Structural unemployment is painful, especially for older workers for whom the best available option might be to retire early or take a lower-skilled, lower-paying job.

Types of Unemployment

- ***Cyclical Unemployment:*** The higher than normal unemployment at a business cycle trough and the lower than normal unemployment at a business cycle peak is called cyclical unemployment

A worker who is laid off because the economy is in a recession and who gets rehired some months later when the expansion begins experiences cyclical unemployment

“Natural” Unemployment

- ***Natural Unemployment:*** Natural unemployment is the unemployment that arises from frictions and structural changes when there is no cyclical unemployment, meaning all the unemployment is frictional and structural

Natural unemployment as a percentage of the labour force is called the natural unemployment rate

Full Employment

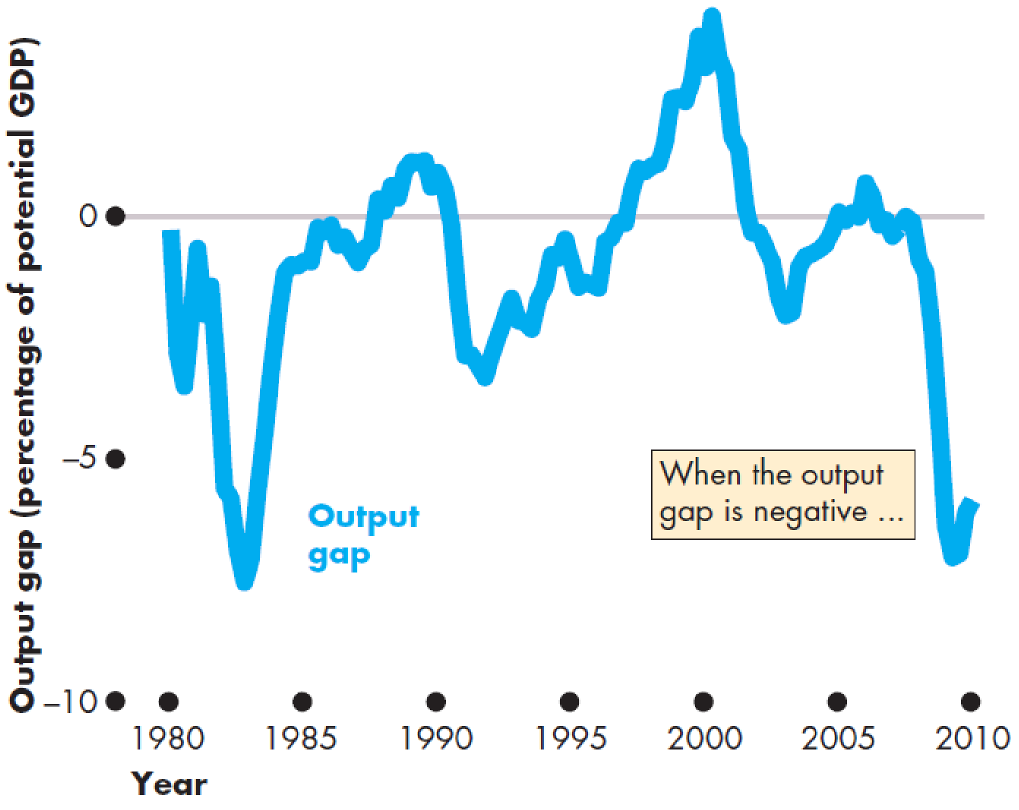
- ***Full employment*** is defined as a situation in which the unemployment rate equals the natural unemployment rate

- Factors that influences Natural Unemployment Rate:
  - Age Distribution of the Population
  - Scale of Structural Change
  - Real Wage Rate
    - Minimum Wage
    - Efficiency Wage
  - Unemployment Benefits

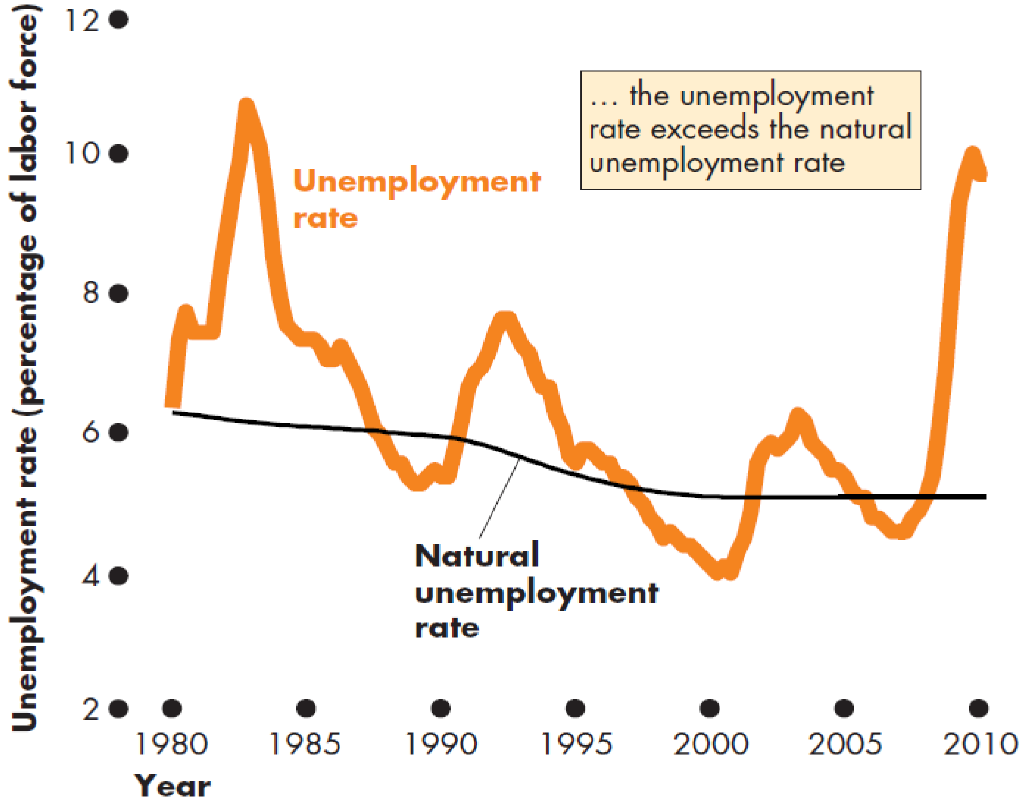
Full Employment and Potential GDP

- The quantity of real GDP at full employment is potential GDP
- Over the business cycle, real GDP fluctuates around potential GDP. The gap between real GDP and potential GDP is called the **output gap**
- As the output gap fluctuates over the business cycle, the unemployment rate fluctuates around the natural unemployment rate

Full Employment and Potential GDP



(a) Output gap



(b) Unemployment rate

Price Level, Inflation and Deflation

- ***Price Level:*** The average level of prices, and the value of money
- A persistently rising price level is called ***inflation***
- A persistently falling price level is called ***deflation***

Question

HOW WOULD YOU THINK AI WILL AFFECT THE LABOUR MARKET?

SHOULD WE BE WORRIED?